ADAM v32: Power Connection & Management Guide

A stable robot requires clean, isolated power. If the servos share power directly through the Raspberry Pi or ESP32-CAM pins, the motors will cause voltage spikes (brownouts) that will crash the AI, drop the Wi-Fi, or cause the audio to stutter.

This guide outlines a **Central 5V Rail** architecture to safely power the Body and Head components.

⚡ 1. The Power Source & The Golden Rule

- **The Battery:** A standard 2S LiPo (7.4V) or 18650 battery pack.
- **The Buck Converter:** You must use a **5V Step-Down Buck Converter** rated for at least **3 Amps** (e.g., LM2596 or MP1584).
- **THE GOLDEN RULE: All grounds must be connected together.** The Pi, ESP32-CAM, Pico, Amplifier, Servos, and Buck Converter MUST share a common Ground (GND). If they don't, I2S audio and UART serial communication will fail.

🏢 2. Body Power Distribution (The Base)

The 5V output from the Buck Converter creates your **Main 5V Rail**.

Component	Power Pin	Connects To	Voltage Required	Notes
Raspberry Pi Zero 2W	5V (Pin 2 or 4)	**Main 5V Rail**	5V	Powers the brain.
MAX98357A (Amp)	VIN	**Main 5V Rail**	5V	Needs high current for the 3W speaker.
MG90S Pan Servo	VCC (Red Wire)	**Main 5V Rail**	5V	**NEVER power from Pi.** Connect directly to rail.
RCWL-0516 Radar	VIN	**Main 5V Rail**	5V	Contains its own internal 3.3V regulator.

INMP441 Mics (x2)	VDD	**Pi 3.3V (Pin 1 or 17)**	3.3V	⚠️ **CRITICAL:** Mics must use 3.3V from the Pi. Using 5V will fry the Pi's I2S data pins.

🌉 3. The Neck Joint (Power Passthrough)

To power the head, run two appropriately thick wires (e.g., 22 AWG) from the **Main 5V Rail** in the body, straight up through the mechanical neck, into the head.

1. **Head 5V Wire**
2. **Head GND Wire**

(If using Flow 2 "Full Hardwired" from the Blueprint, you will also pass the Pi TX and Pi RX wires here).

🤖 4. Head Power Distribution (The Moving Part)

Inside the head, split the 5V and GND wires you passed through the neck to create a **Head 5V Rail**.

Component	Power Pin	Connects To	Voltage Required	Notes
ESP32-CAM	5V Input	**Head 5V Rail**	5V	The ESP32-CAM has a built-in 3.3V regulator for its logic.
Raspberry Pi Pico	VSYS (Pin 39)	**Head 5V Rail**	5V	Do not use VBUS if powering via battery. VSYS is safest.
MG90S Tilt Servo	VCC (Red Wire)	**Head 5V Rail**	5V	**NEVER power from ESP/Pico.** Connect directly to rail.

ILI9341 TFT Display	VCC & LED	**Pico 3V3(OUT) Pin 36**	3.3V	⚠️ **CRITICAL:** Feeding 5V to the TFT logic can fry it. Use the Pico's clean 3.3V output.
PCF8574T Expander	VCC	**ESP32-CAM 3.3V Pin**	3.3V	Keeps I2C logic levels strictly at 3.3V.
TTP223 Touch (x4)	VCC	**ESP32-CAM 3.3V Pin**	3.3V	Draws very little current. Daisy-chain from the expander's power.

🚨 5. Final Safety Checklist

1. **Servo Isolation:** Check the red wires for both MG90S servos. They must trace back directly to the Buck Converter's 5V output, **not** to a pin on the Raspberry Pi or ESP32-CAM.
2. **Audio Logic Safety:** Check the VDD pins on both INMP441 microphones. They must go to the Raspberry Pi's 3.3V pin.
3. **Display Logic Safety:** Check the VCC and LED pins on the TFT Display. They must go to the Raspberry Pi Pico's 3.3V pin.
4. **Common Ground:** Use a multimeter in continuity mode. Touch the metal shielding of the Pi's USB port, the metal shielding of the ESP32-CAM's SD slot, and the GND pin of the Amplifier. They should all beep (showing they share the same ground).